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## polymorphism.md

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# How to serialize properties of derived classes with System.Text.Json

In this article, you learn how to serialize properties of derived classes with the `System.Text.Json` namespace.

## Serialize properties of derived classes

Beginning with .NET 7, `System.Text.Json` supports polymorphic type hierarchy serialization and deserialization with attribute annotations.

| Attribute                                                                         | Description                                                                                                                                                                                                |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>&lt;xref:System.Text.Json.Serialization.JsonDerivedTypeAttribute&gt;</code> | When placed on a type declaration, indicates that the specified subtype should be opted into polymorphic serialization. It also exposes the ability to specify a type discriminator.                       |
| <code>&lt;xref:System.Text.Json.Serialization.JsonPolymorphicAttribute&gt;</code> | When placed on a type declaration, indicates that the type should be serialized polymorphically. It also exposes various options to configure polymorphic serialization and deserialization for that type. |

For example, suppose you have a `WeatherForecastBase` class and a derived class `WeatherForecastWithCity`:

```
:::code language="csharp" source="snippets/how-to/csharp/WeatherForecast.cs"
id="WFB"::: :::code language="vb" source="snippets/how-to/vb/WeatherForecast.vb"
id="WFB":::
```

```
:::code language="csharp" source="snippets/how-to/csharp/WeatherForecast.cs"
id="WFWC"::: :::code language="vb" source="snippets/how-to/vb/WeatherForecast.vb"
id="WFWC":::
```

And suppose the type argument of the `Serialize<TValue>` method at compile time is `WeatherForecastBase`:

```
:::code language="csharp" source="snippets/how-to/csharp/SerializePolymorphic.cs"
id="SerializeWithAttributeAnnotations"::: :::code language="vb" source="snippets/how-
to/vb/SerializePolymorphic.vb" id="SerializeWithAttributeAnnotations":::
```

In this scenario, the `City` property is serialized because the `weatherForecastBase` object is actually a `WeatherForecastWithCity` object. This configuration enables polymorphic

serialization for `WeatherForecastBase`, specifically when the runtime type is `WeatherForecastWithCity`:

```
{
  "City": "Milwaukee",
  "Date": "2022-09-26T00:00:00-05:00",
  "TemperatureCelsius": 15,
  "Summary": "Cool"
}
```

While round-tripping of the payload as `WeatherForecastBase` is supported, it won't materialize as a runtime type of `WeatherForecastWithCity`. Instead, it will materialize as a runtime type of `WeatherForecastBase`:

```
WeatherForecastBase value = JsonSerializer.Deserialize<WeatherForecastBase>("""
{
  "City": "Milwaukee",
  "Date": "2022-09-26T00:00:00-05:00",
  "TemperatureCelsius": 15,
  "Summary": "Cool"
}
""");

Console.WriteLine(value is WeatherForecastWithCity); // False
```

```
Dim value As WeatherForecastBase = JsonSerializer.Deserialize(@"
{
  "City": "Milwaukee",
  "Date": "2022-09-26T00:00:00-05:00",
  "TemperatureCelsius": 15,
  "Summary": "Cool"
}")

Console.WriteLine(value is WeatherForecastWithCity) // False
```

The following section describes how to add metadata to enable round-tripping of the derived type.

## Polymorphic type discriminators

---

To enable polymorphic deserialization, you must specify a type discriminator for the derived class:

```

[JsonDerivedType(typeof(WeatherForecastBase), typeDiscriminator: "base")]
[JsonDerivedType(typeof(WeatherForecastWithCity), typeDiscriminator: "withCity")]
public class WeatherForecastBase
{
    public DateTimeOffset Date { get; set; }
    public int TemperatureCelsius { get; set; }
    public string? Summary { get; set; }
}

public class WeatherForecastWithCity : WeatherForecastBase
{
    public string? City { get; set; }
}

```

```

<JsonDerivedType(GetType(WeatherForecastBase), "base")>
<JsonDerivedType(GetType(WeatherForecastWithCity), "withCity")>
Public Class WeatherForecastBase
    Public Property [Date] As DateTimeOffset
    Public Property TemperatureCelsius As Integer
    Public Property Summary As String
End Class

Public Class WeatherForecastWithCity
    Inherits WeatherForecastBase
    Public Property City As String
End Class

```

With the added metadata, specifically, the type discriminator, the serializer can serialize and deserialize the payload as the `WeatherForecastWithCity` type from its base type `WeatherForecastBase`. Serialization emits JSON along with the type discriminator metadata:

```

WeatherForecastBase weather = new WeatherForecastWithCity
{
    City = "Milwaukee",
    Date = new DateTimeOffset(2022, 9, 26, 0, 0, 0, TimeSpan.FromHours(-5)),
    TemperatureCelsius = 15,
    Summary = "Cool"
}
var json = JsonSerializer.Serialize<WeatherForecastBase>(weather, options);
Console.WriteLine(json);
// Sample output:
// {
//     "$type" : "withCity",
//     "City": "Milwaukee",
//     "Date": "2022-09-26T00:00:00-05:00",
//     "TemperatureCelsius": 15,
//     "Summary": "Cool"
// }

```

```

Dim weather As WeatherForecastBase = New WeatherForecastWithCity With
{
    .City = "Milwaukee",
    .[Date] = New DateTimeOffset(2022, 9, 26, 0, 0, 0, TimeSpan.FromHours(-5)),
    .TemperatureCelsius = 15,
    .Summary = "Cool"
}
Dim json As String = JsonSerializer.Serialize(weather, options)
Console.WriteLine(json)
' Sample output:
' {
'     "$type" : "withCity",
'     "City": "Milwaukee",
'     "Date": "2022-09-26T00:00:00-05:00",
'     "TemperatureCelsius": 15,
'     "Summary": "Cool"
' }

```

With the type discriminator, the serializer can deserialize the payload polymorphically as `WeatherForecastWithCity`:

```

WeatherForecastBase value = JsonSerializer.Deserialize<WeatherForecastBase>(json);
Console.WriteLine(value is WeatherForecastWithCity); // True

```

```
Dim value As WeatherForecastBase = JsonSerializer.Deserialize(json)
Console.WriteLine(value is WeatherForecastWithCity) // True
```

<!--markdownlint-disable MD031-->

[!NOTE] By default, the `$type` discriminator must be placed at the start of the JSON object, grouped together with other metadata properties like `$id` and `$ref`. If you're reading data off an external API that places the `$type` discriminator in the middle of the JSON object, set

<xref:System.Text.Json.JsonSerializerOptions.AllowOutOfOrderMetadataProperties?displayProperty=nameWithType> to `true`: ``csharp JsonSerializerOptions options = new() { AllowOutOfOrderMetadataProperties = true }; JsonSerializer.Deserialize<Base>("""{"Name": "Name", "\$type": "derived"}""", options); `` Be careful when you enable this flag, as it might result in over-buffering (and out-of-memory failures) when performing streaming deserialization of very large JSON objects.

<!--markdownlint-enable MD031-->

## Mix and match type discriminator formats

Type discriminator identifiers are valid in either `string` or `int` forms, so the following is valid:

```
[JsonDerivedType(typeof(WeatherForecastWithCity), 0)]
[JsonDerivedType(typeof(WeatherForecastWithTimeSeries), 1)]
[JsonDerivedType(typeof(WeatherForecastWithLocalNews), 2)]
public class WeatherForecastBase { }

var json = JsonSerializer.Serialize<WeatherForecastBase>(new
WeatherForecastWithTimeSeries());
Console.WriteLine(json);
// Sample output:
// {
//   "$type" : 1,
//   Omitted for brevity...
// }
```

```

<JsonDerivedType(GetType(WeatherForecastWithCity), 0)>
<JsonDerivedType(GetType(WeatherForecastWithTimeSeries), 1)>
<JsonDerivedType(GetType(WeatherForecastWithLocalNews), 2)>
Public Class WeatherForecastBase
End Class

Dim json As String = JsonSerializer.Serialize(Of WeatherForecastBase)(New
WeatherForecastWithTimeSeries())
Console.WriteLine(json)
' Sample output:
' {
'   "$type" : 1,
'   Omitted for brevity...
' }

```

While the API supports mixing and matching type discriminator configurations, it's not recommended. The general recommendation is to use either all `string` type discriminators, all `int` type discriminators, or no discriminators at all. The following example shows how to mix and match type discriminator configurations:

```

[JsonDerivedType(typeof(ThreeDimensionalPoint), typeDiscriminator: 3)]
[JsonDerivedType(typeof(FourDimensionalPoint), typeDiscriminator: "4d")]
public class BasePoint
{
    public int X { get; set; }
    public int Y { get; set; }
}

public class ThreeDimensionalPoint : BasePoint
{
    public int Z { get; set; }
}

public sealed class FourDimensionalPoint : ThreeDimensionalPoint
{
    public int W { get; set; }
}

```

```

<JsonDerivedType(GetType(ThreeDimensionalPoint), 3)>
<JsonDerivedType(GetType(FourDimensionalPoint), "4d")>
Public Class BasePoint
    Public Property X As Integer
    Public Property Y As Integer
End Class

Public Class ThreeDimensionalPoint
    Inherits BasePoint
    Public Property Z As Integer
End Class

Public NotInheritable Class FourDimensionalPoint
    Inherits ThreeDimensionalPoint
    Public Property W As Integer
End Class

```

In the preceding example, the `BasePoint` type doesn't have a type discriminator, while the `ThreeDimensionalPoint` type has an `int` type discriminator, and the `FourDimensionalPoint` has a `string` type discriminator.

[!IMPORTANT] For polymorphic serialization to work, the type of the serialized value should be that of the polymorphic base type. This includes using the base type as the generic type parameter when serializing root-level values, as the declared type of serialized properties, or as the collection element in serialized collections.

```

using System.Text.Json;
using System.Text.Json.Serialization;

PerformRoundTrip<BasePoint>();
PerformRoundTrip<ThreeDimensionalPoint>();
PerformRoundTrip<FourDimensionalPoint>();

static void PerformRoundTrip<T>() where T : BasePoint, new()
{
    var json = JsonSerializer.Serialize<BasePoint>(new T());
    Console.WriteLine(json);

    BasePoint? result = JsonSerializer.Deserialize<BasePoint>(json);
    Console.WriteLine($"result is {typeof(T)}; // {result is T}");
    Console.WriteLine();
}

// Sample output:
// { "X": 541, "Y": 503 }
// result is BasePoint; // True
//
// { "$type": 3, "Z": 399, "X": 835, "Y": 78 }
// result is ThreeDimensionalPoint; // True
//
// { "$type": "4d", "W": 993, "Z": 427, "X": 508, "Y": 741 }
// result is FourDimensionalPoint; // True

```

```
Imports System.Text.Json
Imports System.Text.Json.Serialization

Module Program
    Sub Main()
        PerformRoundTrip(Of BasePoint)()
        PerformRoundTrip(Of ThreeDimensionalPoint)()
        PerformRoundTrip(Of FourDimensionalPoint)()
    End Sub

    Private Sub PerformRoundTrip(Of T As {BasePoint, NewNew T())
        Console.WriteLine(json)

        Dim result As BasePoint = JsonSerializer.Deserialize(Of BasePoint)(json)
        Console.WriteLine($"result is {GetType(T)}; // {TypeOf result Is T}")
        Console.WriteLine()
    End Sub
End Module

' Sample output:
' { "X": 649, "Y": 754 }
' result is BasePoint; // True
'
' { "$type": 3, "Z": 247, "X": 814, "Y": 56 }
' result is ThreeDimensionalPoint; // True
'
' { "$type": "4d", "W": 427, "Z": 193, "X": 112, "Y": 935 }
' result is FourDimensionalPoint; // True
```

## Customize the type discriminator name

The default property name for the type discriminator is `$type`. To customize the property name, use the `<xref:System.Text.Json.Serialization.JsonPolymorphicAttribute>` as shown in the following example:

```
[JsonPolymorphic(TypeDiscriminatorPropertyName = "$discriminator")]
[JsonDerivedType(typeof(ThreeDimensionalPoint), typeDiscriminator: "3d")]
public class BasePoint
{
    public int X { get; set; }
    public int Y { get; set; }
}

public sealed class ThreeDimensionalPoint : BasePoint
{
    public int Z { get; set; }
}
```

```
<JsonPolymorphic(TypeDiscriminatorPropertyName:="$discriminator")>
<JsonDerivedType(GetType(ThreeDimensionalPoint), "3d")>
Public Class BasePoint
    Public Property X As Integer
    Public Property Y As Integer
End Class

Public Class ThreeDimensionalPoint
    Inherits BasePoint
    Public Property Z As Integer
End Class
```

In the preceding code, the `JsonPolymorphic` attribute configures the `TypeDiscriminatorPropertyName` to the `"$discriminator"` value. With the type discriminator name configured, the following example shows the `ThreeDimensionalPoint` type serialized as JSON:

```
BasePoint point = new ThreeDimensionalPoint { X = 1, Y = 2, Z = 3 };
var json = JsonSerializer.Serialize<BasePoint>(point);
Console.WriteLine(json);
// Sample output:
// { "$discriminator": "3d", "X": 1, "Y": 2, "Z": 3 }
```

```
Dim point As BasePoint = New ThreeDimensionalPoint With { .X = 1, .Y = 2, .Z = 3 }
Dim json As String = JsonSerializer.Serialize(Of BasePoint)(point)
Console.WriteLine(json)
' Sample output:
' { "$discriminator": "3d", "X": 1, "Y": 2, "Z": 3 }
```

[!TIP] Avoid using a `<xref:System.Text.Json.Serialization.JsonPolymorphicAttribute.TypeDiscriminatorPropertyName?displayProperty=nameWithType>` that conflicts with a property in your type hierarchy.

## Handle unknown derived types

To handle unknown derived types, you must opt in to such support using an annotation on the base type. Consider the following type hierarchy:

```
[JsonDerivedType(typeof(ThreeDimensionalPoint))]  
public class BasePoint  
{  
    public int X { get; set; }  
    public int Y { get; set; }  
}  
  
public class ThreeDimensionalPoint : BasePoint  
{  
    public int Z { get; set; }  
}  
  
public class FourDimensionalPoint : ThreeDimensionalPoint  
{  
    public int W { get; set; }  
}
```

```
<JsonDerivedType(GetType(ThreeDimensionalPoint))>  
Public Class BasePoint  
    Public Property X As Integer  
    Public Property Y As Integer  
End Class  
  
Public Class ThreeDimensionalPoint  
    Inherits BasePoint  
    Public Property Z As Integer  
End Class  
  
Public NotInheritable Class FourDimensionalPoint  
    Inherits ThreeDimensionalPoint  
    Public Property W As Integer  
End Class
```

Since the configuration does not explicitly opt-in support for `FourDimensionalPoint`, attempting to serialize instances of `FourDimensionalPoint` as `BasePoint` will result in a runtime exception:

```
JsonSerializer.Serialize<BasePoint>(new FourDimensionalPoint()); // throws  
NotSupportedException
```

```
JsonSerializer.Serialize(Of BasePoint)(New FourDimensionalPoint()) ' throws  
NotSupportedException
```

You can change the default behavior by using the `<xref:System.Text.Json.Serialization.JsonUnknownDerivedTypeHandling>` enum, which can be specified as follows:

```
[JsonPolymorphic(  
    UnknownDerivedTypeHandling = JsonUnknownDerivedTypeHandling.FallBackToBaseType)]  
[JsonDerivedType(typeof(ThreeDimensionalPoint))]  
public class BasePoint  
{  
    public int X { get; set; }  
    public int Y { get; set; }  
}  
  
public class ThreeDimensionalPoint : BasePoint  
{  
    public int Z { get; set; }  
}  
  
public class FourDimensionalPoint : ThreeDimensionalPoint  
{  
    public int W { get; set; }  
}
```

```

<JsonPolymorphic(
    UnknownDerivedTypeHandling:=JsonUnknownDerivedTypeHandling.FallBackToBaseType)>
<JsonDerivedType(GetType(ThreeDimensionalPoint))>
Public Class BasePoint
    Public Property X As Integer
    Public Property Y As Integer
End Class

Public Class ThreeDimensionalPoint
    Inherits BasePoint
    Public Property Z As Integer
End Class

Public NotInheritable Class FourDimensionalPoint
    Inherits ThreeDimensionalPoint
    Public Property W As Integer
End Class

```

Instead of falling back to the base type, you can use the `FallBackToNearestAncestor` setting to fall back to the contract of the nearest declared derived type:

```

[JsonPolymorphic(
    UnknownDerivedTypeHandling =
    JsonUnknownDerivedTypeHandling.FallBackToNearestAncestor)]
[JsonDerivedType(typeof(BasePoint)))]
public interface IPoint { }

public class BasePoint : IPoint { }

public class ThreeDimensionalPoint : BasePoint { }

```

```

<JsonPolymorphic(
    UnknownDerivedTypeHandling =
    JsonUnknownDerivedTypeHandling.FallBackToNearestAncestor)>
<JsonDerivedType(GetType(BasePoint))>
Public Interface IPoint
End Interface

Public Class BasePoint
    Inherits IPoint
End Class

Public Class ThreeDimensionalPoint
    Inherits BasePoint
End Class

```

With a configuration like the preceding example, the `ThreeDimensionalPoint` type will be serialized as `BasePoint` :

```

// Serializes using the contract for BasePoint
JsonSerializer.Serialize<IPoint>(new ThreeDimensionalPoint());

```

```

' Serializes using the contract for BasePoint
JsonSerializer.Serialize(Of IPoint)(New ThreeDimensionalPoint())

```

However, falling back to the nearest ancestor admits the possibility of "diamond" ambiguity. Consider the following type hierarchy as an example:

```

[JsonPolymorphic(
    UnknownDerivedTypeHandling =
    JsonUnknownDerivedTypeHandling.FallBackToNearestAncestor)]
[JsonDerivedType(typeof(BasePoint))]
[JsonDerivedType(typeof(IPointWithTimeSeries))]
public interface IPoint { }

public interface IPointWithTimeSeries : IPoint { }

public class BasePoint : IPoint { }

public class BasePointWithTimeSeries : BasePoint, IPointWithTimeSeries { }

```

```

<JsonPolymorphic(

    UnknownDerivedTypeHandling:=JsonUnknownDerivedTypeHandling.FallBackToNearestAncestor)>
<JsonDerivedType(GetType(BasePoint))>
<JsonDerivedType(GetType(IPointWithTimeSeries))>
Public Interface IPoint
End Interface

Public Interface IPointWithTimeSeries
    Inherits IPoint
End Interface

Public Class BasePoint
    Implements IPoint
End Class

Public Class BasePointWithTimeSeries
    Inherits BasePoint
    Implements IPointWithTimeSeries
End Class

```

In this case, the `BasePointWithTimeSeries` type could be serialized as either `BasePoint` or `IPointWithTimeSeries` since they are both direct ancestors. This ambiguity will cause the `<xref:System.NotSupportedException>` to be thrown when attempting to serialize an instance of `BasePointWithTimeSeries` as `IPoint`.

```

// throws NotSupportedException
JsonSerializer.Serialize<IPoint>(new BasePointWithTimeSeries());

```

```

' throws NotSupportedException
JsonSerializer.Serialize(Of IPoint)(New BasePointWithTimeSeries())

```

## Configure polymorphism with the contract model

For use cases where attribute annotations are impractical or impossible (such as large domain models, cross-assembly hierarchies, or hierarchies in third-party dependencies), to configure polymorphism use the [contract model](#). The contract model is a set of APIs that can be used to configure polymorphism in a type hierarchy by creating a custom `<xref:System.Text.Json.Serialization.Metadata.DefaultJsonTypeInfoResolver>` subclass that dynamically provides polymorphic configuration per type, as shown in the following example:

```

public class PolymorphicTypeResolver : DefaultJsonTypeInfoResolver
{
    public override JsonTypeInfo GetTypeInfo(Type type, JsonSerializerOptions options)
    {
        JsonTypeInfo jsonTypeInfo = base.GetTypeInfo(type, options);

        Type basePointType = typeof(BasePoint);
        if (jsonTypeInfo.Type == basePointType)
        {
            jsonTypeInfo.PolymorphismOptions = new JsonPolymorphismOptions
            {
                TypeDiscriminatorPropertyName = "$point-type",
                IgnoreUnrecognizedTypeDiscriminators = true,
                UnknownDerivedTypeHandling =
                JsonUnknownDerivedTypeHandling.FailSerialization,
                DerivedTypes =
                {
                    new JsonDerivedType(typeof(ThreeDimensionalPoint), "3d"),
                    new JsonDerivedType(typeof(FourDimensionalPoint), "4d")
                }
            };
        }

        return jsonTypeInfo;
    }
}

```

```

Public Class PolymorphicTypeResolver
    Inherits DefaultJsonTypeInfoResolver

    Public Overrides Function GetTypeInfo(
        ByVal type As Type,
        ByVal options As JsonSerializerOptions) As JsonTypeInfo

        Dim jsonTypeInfo As JsonTypeInfo = MyBase.GetTypeInfo(type, options)
        Dim basePointType As Type = GetType(BasePoint)

        If jsonTypeInfo.Type = basePointType Then
            jsonTypeInfo.PolymorphismOptions = New JsonPolymorphismOptions With {
                .TypeDiscriminatorPropertyName = "$point-type",
                .IgnoreUnrecognizedTypeDiscriminators = True,
                .UnknownDerivedTypeHandling =
                    JsonUnknownDerivedTypeHandling.FailSerialization
            }
            jsonTypeInfo.PolymorphismOptions.DerivedTypes.Add(
                New JsonDerivedType(GetType(ThreeDimensionalPoint), "3d"))
            jsonTypeInfo.PolymorphismOptions.DerivedTypes.Add(
                New JsonDerivedType(GetType(FourDimensionalPoint), "4d"))
        End If

        Return jsonTypeInfo
    End Function
End Class

```

## Additional polymorphic serialization details

- Polymorphic serialization supports derived types that have been explicitly opted in via the `<xref:System.Text.Json.Serialization.JsonDerivedTypeAttribute>`. Undeclared types will result in a runtime exception. The behavior can be changed by configuring the `<xref:System.Text.Json.Serialization.JsonPolymorphicAttribute.UnknownDerivedTypeHandling?displayProperty=nameWithType>` property.
- Polymorphic configuration specified in derived types is not inherited by polymorphic configuration in base types. The base type must be configured independently.
- Polymorphic hierarchies are supported for both `interface` and `class` types.
- Polymorphism using type discriminators is only supported for type hierarchies that use the default converters for objects, collections, and dictionary types.
- Polymorphism is supported in metadata-based source generation, but not fast-path source generation.

## See also

- [System.Text.Json overview](#)
- [How to serialize and deserialize JSON](#)

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